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# Accelerating the Miller–Tucker–Zemlin model for the asymmetric traveling salesman problem

Giovanni Campuzano[a,*], Carlos Obreque[a], Maichel M. Aguayo[b]

[a] Industrial Engineering Department, Universidad del Bío-Bío, Concepción, Chile
[b] Industrial Engineering Department, Universidad de Concepción, Concepción, Chile

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ABSTRACT

In this article, we present an easy to implement algorithmic approach that improves the computational performance of the Miller–Tucker–Zemlin (MTZ) model for the asymmetric traveling salesman problem (ATSP) by efficiently generating valid inequalities from fractional solutions. Computational experiments show that the proposed approach enhances considerably the performance of MTZ-based formulations reported in the literature. By adding facet-defining inequalities of the underline ATSP-polytope, the number of nodes in the branch-and-bound tree is drastically reduced, and the convergence of the MTZ-type formulations is accelerated. We also extend this idea to solve the multiple asymmetric traveling salesman problem (mATSP). This approach can help practitioners to solve real-life problems to near optimality using a standard optimization solver and may be useful to solve a variety of routing problems that use MTZ-type of subtour elimination constraints.

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## 1. Introduction

The traveling salesman problem (TSP) is one of the most well-known combinational optimization problems studied in the operations research literature. It consists of determining a tour that starts and ends at a given base node after having visited a set of nodes exactly once while minimizing the total distance. Applications of the TSP to real-world problems consider computer wiring, wallpaper cutting, and crystallography, among many others (Reinelt, 1994). The TSP is classified into symmetric or asymmetric, based on whether the distance or cost to travel between any two nodes is symmetric or asymmetric, respectively. This paper deals with ATSP and its extension, the multiple ATSP (mATSP), where more than one salesman is used in the solution.

The ATSP has several applications to real-world problems, especially in city logistics and scheduling. In city logistics, ATSP can be used to determine optimal routes for mobile waste electrical and electronic equipment collection (Nowakowski, Szwarc, & Boryczka, 2018), design street sweeping routes (Blazquez, Beghelli, & Meneses, 2012), and determine the number of trains running on a given network per unit time (Gonsalves & Shiozaki, 2015). In scheduling, the ATSP can be used to determine optimal schedules in a single machine problem with sequence-dependent setup times with makespan objective and for the no-wait flow-shop scheduling problem with makespan objective (Mucha & Sviridenko, 2013).

The mATSP also has applications in many real life-situations. It can be used to design a global navigation satellite system surveying networks (Saleh & Chelouah, 2004), solve the print press scheduling problem (Gorenstein, 1970), model maintenance network (Kota & Jarmai, 2015), solve the school bus routing problem (Angel, Caudle, Noonan, & Whinston, 1972; Ellegood, Solomon, North, & Campbell, 2019), and model scheduling problems (Jiang, Wan, & Peng, 2020).

ATSP and mATSP are widely applied in industrial situations, and algorithms solving these problems are embedded in decision support systems (DSS) that support managerial decision-making. All these facts have encouraged us to improve the MTZ model for the ATSP and mATSP to help practitioners to solve real-life problems to near optimality using a standard optimization solver.

The main contribution of this paper is an easy to implement algorithmic approach that improves the computational performance of the MTZ model for ATSP and mATSP. We efficiently generate subtour elimination constraints (SEC) (facet-defining inequalities) as valid inequalities from fractional solutions while the original formulation is solved. This approach enhances the performance of MTZ-based formulation considerably and obtains competitive solutions in relation to those reported by a recent state-of-the-art metaheuristic.

* Corresponding author
*E-mail addresses:* gcampuzano@ubiobio.cl (G. Campuzano), cobreque@ubiobio.cl (C. Obreque), maichelaguayo@udec.cl (M.M. Aguayo).

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**Table 1**
Summary of the existing enhancement of the MTZ-SECs.

| Authors | Approach to enhance the MTZ-SECs |
|---|---|
| Desrochers and Laporte (1991) | Lifting of the MTZ-SECs |
| Gouveia and Pires (1999) | Disaggregation of the MTZ-SECs |
| Sherali and Driscoll (2001) | Reformulation of a nonlinear restatement of the MTZ-SECs |
| Gouveia and Pires (2001) | Generalizations of the disaggregated MTZ-SECs |
| Bektaş & Gouveia, 2014 | Generalizations of the MTZ and DL inequalities |
| Sawik (2016) | Introduction of boundary constraints in the MTZ-SECs |
| Our paper | Introduction of valid inequalities in the MTZ-SECs |

The rest of the article is structured as follows. In Section 2, we present a review of related papers improving MTZ model for ATSP. In Section 3, we define the problem and present mathematical formulations reported for both ATSP and mATSP. In Section 4, we describe the proposed algorithmic approach to accelerate the MTZ-type formulations, while in Section 5, we present the computation results obtained by the proposed approach and compare it with existing formulations reported in the literature on several asymmetric TSPLIB instances involving between 17-443 nodes. Concluding remarks are stated in Section 6.

## 2. Literature review

The ATSP has been around for a long time; however, it is still an active research area. Roberti and Toth (2012) present the most effective mathematical models and exact algorithms proposed for the ATSP. Since then, new compact formulations for the ATSP are reported in Bektaş & Gouveia, 2014, Gouveia, Pesneau, Ruthmair, and Santos (2018), and Balma, Salem, Mrad, and Ladhari (2018); exact algorithms are introduced in Aguayo, Sarin, and Sherali (2018); and heuristic and metaheuristic algorithms are reported in Osaba, Yang, Diaz, Lopez-Garcia, and Carballedo (2016), Nagata and Soler (2012), Basu, Sharma, and Ghosh (2017), Osaba, Ser, Sadollah, Bilbao, and Camacho (2018), and Boryczka and Szwarc (2019a,b).

Several authors have proposed extensions and enhancement of the MTZ-SECs. Desrochers and Laporte (1991) lift the MTZ-SECs into facets of the underlying ATSP polytope and generalize them to several vehicle routing problems. Gouveia and Pires (1999) propose a stronger version of the MTZ-SECs by disaggregating the information given by the node variables. Sherali and Driscoll (2001) introduce a tighter version of the MTZ-SECs by applying the Reformulation-Linearization Technique (RLT) to a nonlinear restatement of the MTZ-SECs. Gouveia and Pires (2001) propose a generalization of the disaggregated MTZ-SECs (Gouveia & Pires, 1999) and derive an augmented multicommodity flow (MCF) formulation obtained by adding to an MCF model a disaggregated version of the DL inequalities with a set of linking constraints. Bektaş & Gouveia, 2014 generalize the underlying idea of the MTZ and DL-SECs implying CLIQUE constraints for three-node subsets. Sawik (2016) enhance the MTZ-SECs by introducing additional constraints to the initial formulation that account for ordering of boundary nodes as well as intermediate nodes in the salesman tour.

Contrary to the previous ideas to enhance the MTZ-SECs, in this article, we propose an easy to implement algorithmic approach to accelerate the MTZ-SECs by efficiently generating valid inequalities from fractional solutions, which is shown to improve the performance of the original and enhanced formulations reported in the literature. The existing and proposed enhancements of the MTZ-SECs are summarized in Table 1.

Concerning mATSP, compact formulations are surveyed in Kara and Bektas (2006) and Sarin, Sherali, Judd, and Tsai (2014). The latter authors show that the most effective formulations for mATSP when solved directly on CPLEX are the *MTZ, KB* (Kara & Bektas, 2006), and *GG* formulations (Gavish & Graves, 1978), which we consider for comparison in our experiments. Heuristic and metaheuristic algorithms for the problem are reported in Soylu (2015), Essani and Haider (2018), and Jiang et al. (2020). To the best of our knowledge, our proposed approach has been reported in the literature neither for ATSP nor mATSP.

## 3. Problem description

### 3.1. ATSP

The ATSP can be defined as follows: given a complete graph with vertex set $N$ in which node 1 denotes the base and an asymmetric distance matrix $[c_{ij}]$, $i, j \in N$, determines an optimal tour that starts and ends at the base node after having visited node $i$ exactly once, $\forall i \in N$, while minimizing the total distance traveled.

The basic formulation for the ATSP is as follows:

$$\textbf{ATSP:}\ \text{Minimize} \sum_{i\in N}\sum_{j\in N, j\neq i} c_{ij}x_{ij} \tag{1}$$

subject to :

$$\sum_{i\in N, j\neq i} x_{ij} = 1, \quad \forall j \in N \tag{2}$$

$$\sum_{j\in N, j\neq i} x_{ij} = 1, \quad \forall i \in N \tag{3}$$

$$\text{SECs} \tag{4}$$

$$x_{ij} \in \{0,1\}, \quad \forall i, j \in N, \ \ j \neq i. \tag{5}$$

The objective function (1) minimizes the total distance traveled, and constraints (2) and (3) ensure that each node is visited and departed exactly once. Constraints (4) are the subtour elimination constraints (SECs), which can be formulated as an exponential set of constraints (natural formulations) or a polynomial number of restrictions by introducing additional variables (extended formulations). SECs with an exponential set of constraints (DFJ) (Dantzig, Flukerson, & Johnson, 1954) are defined as follows:

$$\sum_{i\in S}\sum_{j\in N\setminus\{S\}} x_{ij} \geq 1, \qquad \forall S \subset N, |S| \geq 2 \tag{6}$$

$$\sum_{i\in S}\sum_{j\in S,\ j\neq i} x_{ij} \leq |S| - 1, \quad \forall S \subset N, |S| \geq 2. \tag{7}$$

Reviews of polynomial-length formulations for the ATSP has been reported in Öncan, Altinel, and Laporte (2009) and Roberti and Toth (2012). In this paper, we focus on the compact formulation MTZ propose in Miller, Tucker, and Zemlin (1960). Velednitsky (2017) presents a short analytical proof that the DFJ

polytope is contained in the MTZ polytope. Let $u_i$ be a variable denoting the position in which node $i$ is visited in the tour; then the MTZ-SECs are as follows:

$$u_i - u_j + (|N|-1)x_{ij} \leq |N|-2, \quad \forall i \in N, j \in N \setminus \{1\}, i \neq j \tag{8}$$

$$1 \leq u_i \leq |N|-1, \qquad \forall i \in N \setminus \{1\}. \tag{9}$$

### 3.2. mATSP

The mATSP can be defined as follows: given a complete graph with vertex set $N$ in which node 1 denotes the base and an asymmetric distance matrix $[c_{ij}]$, $i, j \in N$, determines $m$ optimal tours that start and end at the base node after having visited node $i$ exactly once, $\forall i \in N\backslash\{1\}$, while minimizing the total distance traveled. The basic formulation for the mATSP is as follows:

$$\textbf{mATSP:} \quad \text{Minimize} \sum_{i\in N} \sum_{j\in N, j\neq i} c_{ij}x_{ij} \tag{10}$$

subject to :

$$\sum_{j\in N, j\neq 1} x_{1j} = m \tag{11}$$

$$\sum_{j\in N, j\neq 1} x_{j1} = m \tag{12}$$

$$\sum_{i\in N, i\neq j} x_{ij} = 1, \quad \forall \in N \setminus \{1\} \tag{13}$$

$$\sum_{j\in N, j\neq i} x_{ij} = 1, \quad \forall i \in N \setminus \{1\} \tag{14}$$

$$\text{SECs} \tag{15}$$

$$x_{ij} \in \{0,1\}, \quad \forall i, j \in N, \ i \neq j. \tag{16}$$

Constraints (11) and (12) force $m$ salesmen to depart and return from node 1, while constraints (13) and (14) assure that each node is visited and departed once, respectively. SECs (6) are valid for mATSP, but SECs (7) need to be redefined as follows:

$$\sum_{i\in S} \sum_{j\in S,\ j\neq i} x_{ij} \leq |S|-1, \quad \forall S \subset N \setminus \{1\}, |S| \geq 2 \tag{17}$$

The MTZ-SECs for the mATSP are as follows:

$$u_i - u_j + (|N|-m)x_{ij} \leq |N|-m-1, \quad \forall i \in N, j \in N \setminus \{1\}, i \neq j \tag{18}$$

$$1 \leq u_i \leq |N|-m, \qquad \forall i \in N \setminus \{1\} \tag{19}$$

## 4. Propose approach for ATSP and mATSP

### 4.1. Fractional islands

The weakness of the linear relaxation of the *MTZ*-formulation has been reported in Desrochers and Laporte (1991), Gouveia and Pires (1999), and Velednitsky (2017). In our preliminary investigations, we observe that when MTZ-formulation is solved in standard optimization software, at each node of the branch-and-bound algorithm, it consistently generates disjoint subgraph of nodes. We denote them by "fractional islands." Fig. 1 presents the fractional solutions obtained from the ATSP instance called *br*17 consisting of 17 nodes while solving the original MTZ-based formulation. For illustration purpose, we show 7 out of the 17 nodes. The value of the $x_{ij}$- and $u_i$-variables are shown in this figure. The fractional solutions resulting from solving the MTZ-formulation generate disjoint subgraph of nodes (fractional islands). There are two fractional islands which are identified by sets $S_1 = \{7, 8, 16\}$ and $S_2 = \{1, 2, 11, 13\}$. Given the $x_{ij}$-values, the $u_i$ satisfies the MTZ-SECs.

In this work, we propose to improve the MTZ-linear programming (LP) relaxation by generating valid inequalities (SECs) from fractional solutions obtained during the optimization process, which can cut off fractional solutions and be separated efficiently. Fractional islands appear consistently during the optimization process; e.g., when solving the instance balas200, we observe that fractional islands appear 4103 times. Thus, we intend to add those valid inequalities for all fractional islands that will cut off these solutions from the LP relaxation; and therefore, this will speed-up solution times.

### 4.2. Propose approach

We propose to efficiently separate fractional islands and embed the identified valid inequalities in a commercial solver while solving the original MTZ-formulation. The proposed approach is illustrated in Fig. 2. This approach consists in taking advantage of modern optimization software and is described next.

- **Step 1:** Solve the MIP model (1)–(5) incorporating MTZ-SECs (8) and (9) until an optimal solution is found or a time limit is reached. We use CPLEX 12.8 with default parameters to solve the MIP model. The optimization process is continuously monitored with a CPLEX callback and whenever CPLEX finds a fractional solution to the relaxed problem, go to Step 2.
- **Step 2.** Whenever the CPLEX callback identifies fractional solutions, we use a separation algorithm (Shrinking method) to identify valid inequalities from LP-relaxation values ($x_{ij}$ values). These values are obtained invoking "get relaxation point" from the new generic callback of CPLEX (see details in (IBM, 2019)). Then, go to Step 3.
- **Step 3:** If fractional islands are detected, go to Step 4; otherwise, continue solving the MIP model in Step 1.
- **Step 4.** Generate and add to the MIP model the strong valid inequalities identified. For each set of nodes forming a fractional island that do not satisfy (6) or (7), $S_i$, $i = 1, ..., k$, where $k$ represents the number of subtours detected in the solution, add a valid inequality. Then, go to Step 1.

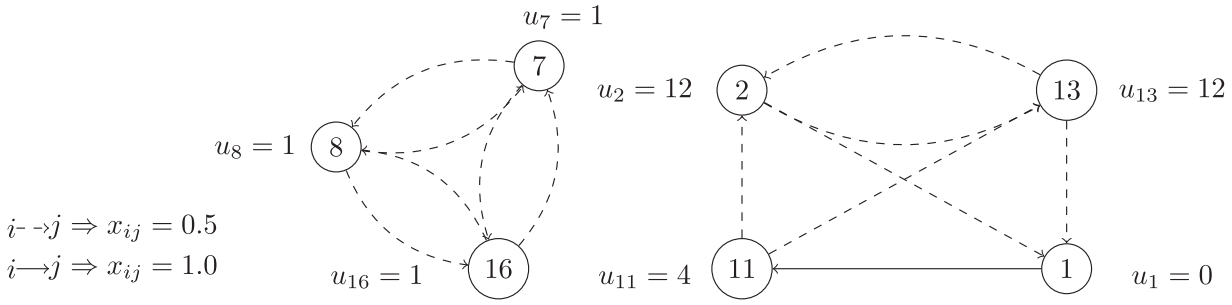
We use the Shrinking method to identify fractional islands (Applegate, Bixby, Chvátal, & Cook, 2001). The set of islands is denoted as $S_i$, $i = 1, ..., k$ such that $N = \bigcup_{i=1}^{k} S_i$. Let $i = 0$ and $N' = N$. Shrinking uses a supernode $S$ and place a node from $N'$ in $S$ to start the search; and then, it adds to $S$ all nodes $v$ such $x_{uv} \neq 0 : u \in S, v \in N' \setminus \{S\}$. Repeat this search every time a node is added to $S$ and until no more nodes can be added. Then, make $i = i + 1$, $S_i = S$, $N' = N' \setminus \{S\}$, and $S = \emptyset$. Repeat this process until $N' = \emptyset$.

Typically, a user will only perform Step 1 to solve ATSP with the classical *MTZ*; however, we propose a simple yet effective idea embedding Steps 2-4 to improve the convergence of the model drastically.

To apply, the algorithmic approach displayed in Fig. 2 to mATSP, In Step 1, we solve the MIP model (10)–(16) with MTZ-SECs (18) and (19), while in Step 4, we use the valid inequalities (6) or (17).

## 5. Computational results

The algorithmic approach was coded in C ++ Microsoft Visual Studio 2017 with CPLEX 12.8 with default parameters except that




$$1 \leq u_i \leq |N| - m, \qquad \forall i \in N \setminus \{1\} \tag{19}$$

**Fig. 1.** Two fractional islands obtained from the ATSP instance *br*17. Sets $S_1 = \{8, 7, 16\}$ and $S_2 = \{1, 2, 11, 13\}$ are identified.

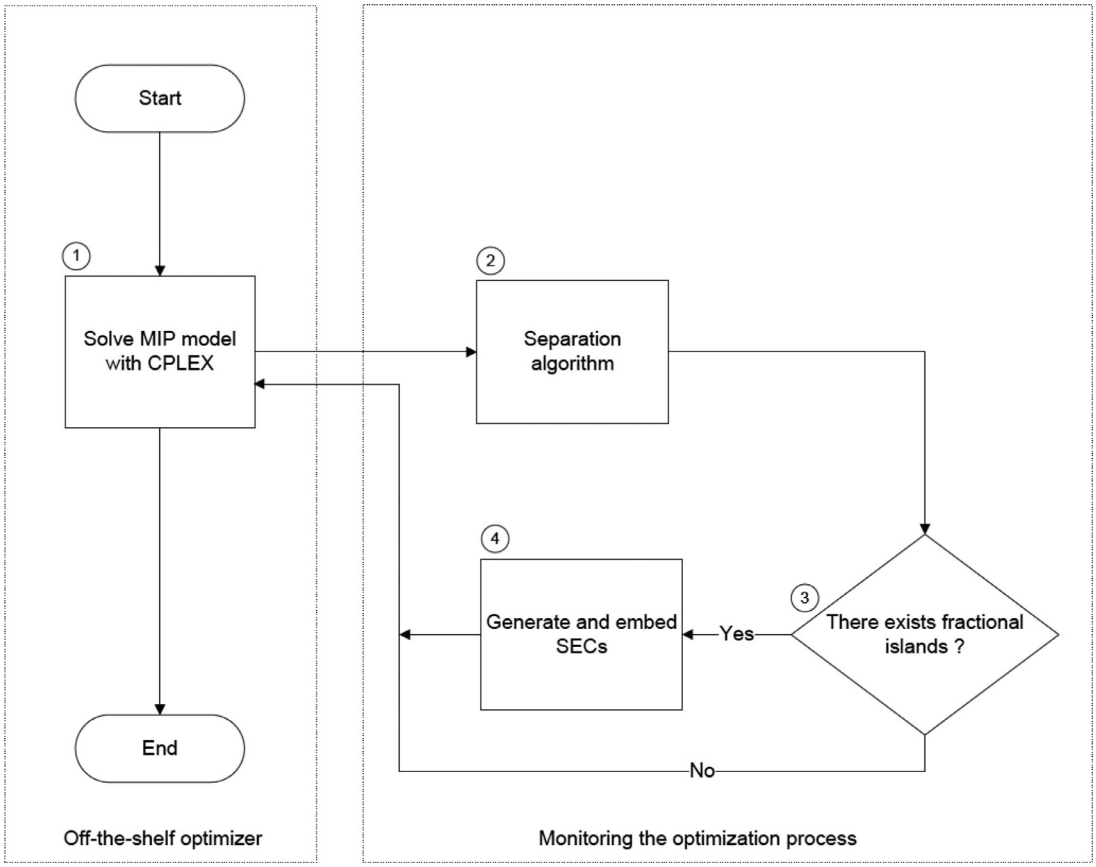



**Fig. 2.** Flow diagram of the proposed approach.

we set the deterministic parallel mode so that multiple runs with the same model at the same parameter settings on the same platform will reproduce the same solution path and results. We use the new CPLEX callbacks, which allow accessing the fractional solutions during the search; and then, to add the generated SECs to the model without affecting its default parameters. All experiments were made in Intel Core i7-4770 3.40 GHz computer having 16 GB of RAM running Windows 10 with a time limit of 3600 seconds. Furthermore, we have used 32 instances involving 17-443 nodes from the TSPLIB, which are shown in Table 2.

### 5.1. Results for ATSP

Table 3 presents a comparison between the proposed accelerated *MTZ* ($MTZ^{acc}$) and the classical *MTZ* (Miller et al., 1960) as well as enhanced *MTZ* formulations reported in the literature; *DL* (Desrochers & Laporte, 1991), *SD* (Sherali & Driscoll, 2001), and *SAWIK_en* and *SAWIK_en−* (Sawik, 2016). We also compare $MTZ^{acc}$ with the *GG*-formulation (Gavish & Graves, 1978), one of the most effective ATSP formulation reported in the literature (Roberti & Toth, 2012). The first column of Table 3 indicates the name of the instance (Instance), while the remaining columns show the time to solve the instance to optimality or the optimality gap reported by CPLEX in terms of percentage (%) in case an optimal solution is not reached. An underlined value indicates the optimality gap and implies that the time for the algorithmic is 3600 s. For example, in Table 3, instance p43y, the formulation MTZ failed to obtain an optimal solution within the time limit; thus, we report 1.29%. Otherwise, this value denotes the computational time to solve the problem to optimality. We highlight the minimum time or gap is

**Table 2**
Set of instances from the TSPLIB involving 17-443 nodes considered in our computational experiments.

| Name | $\|N\|$ | Optimal solution | Name | $\|N\|$ | Optimal solution |
|---|---|---|---|---|---|
| br17 | 17 | 39 | ftv100 | 101 | 1788 |
| ftv33 | 34 | 1286 | balas108 | 108 | 152 |
| ftv35 | 36 | 1473 | ftv110 | 111 | 1958 |
| ftv38 | 39 | 1530 | balas120 | 120 | 286 |
| p43y | 43 | 5620 | ftv120 | 121 | 2166 |
| ftv44 | 45 | 1613 | ftv130 | 131 | 2307 |
| ftv47 | 48 | 1776 | ftv140 | 141 | 2420 |
| ry48p | 48 | 14422 | ftv150 | 151 | 2611 |
| ft53 | 53 | 6905 | balas160 | 160 | 397 |
| ftv55 | 56 | 1608 | ftv160 | 161 | 2683 |
| ftv64 | 65 | 1839 | ftv170 | 171 | 2755 |
| ft70 | 70 | 38673 | balas200 | 200 | 403 |
| ftv70 | 71 | 1950 | rbg323 | 323 | 1326 |
| balas84 | 84 | 199 | rbg358 | 358 | 1163 |
| ftv90 | 91 | 1579 | rbg403 | 403 | 2465 |
| kro124p | 100 | 36230 | rbg443 | 443 | 2720 |

**Table 3**
Comparison of the results obtained by the proposed approach ($MTZ^{acc}$) with those obtained by the *MTZ*, *DL*, *GG*, *SD*, *SAWIK_en*, and *SAWIK_en−* formulations on 32 ATSP instances.

| *Instance* | $MTZ^{acc}$ | *MTZ* | *DL* | *GG* | *SD* | *SAWIK_en* | *SAWIK_en−* |
|---|---|---|---|---|---|---|---|
| br17 | **0.15** | 0.57 | 0.53 | 0.27 | 0.59 | 0.57 | 1.67 |
| ftv33 | **0.27** | 0.53 | 0.28 | 0.78 | 1.64 | 0.80 | 0.93 |
| ftv35 | **0.45** | 0.53 | 0.89 | 0.78 | 2.07 | 0.93 | 1.06 |
| ftv38 | **0.32** | 0.63 | 0.70 | 1.78 | 2.77 | 0.59 | 0.82 |
| p43y | **1.18** | 1.29% | 0.93% | 23.69 | 711.98 | 0.64% | 0.58% |
| ftv44 | 1.11 | **0.90** | 1.33 | 2.16 | 2.84 | 1.23 | 1.98 |
| ftv47 | **0.80** | 1.94 | 3.44 | 6.81 | 20.87 | 4.41 | 3.09 |
| ry48p | **1.01** | 14.88 | 10.69 | 9.97 | 49.01 | 28.33 | 66.01 |
| ft53 | **1.23** | 13.85 | 2.22 | 3.46 | 29.93 | 19.13 | 12.05 |
| ftv55 | **1.89** | 3.62 | 1.89 | 7.97 | 71.24 | 6.45 | 7.80 |
| ftv64 | **2.53** | 8.02 | 7.14 | 19.02 | 103.20 | 10.83 | 10.16 |
| ft70 | **2.01** | 0.10% | 0.05% | 6.52 | 16.48 | 2.98 | 4.34 |
| ftv70 | **3.57** | 7.99 | 9.69 | 13.40 | 171.44 | 14.46 | 15.65 |
| balas84 | **74.29** | 1319.23 | 1102.89 | 454.46 | 10.00% | 1807.99 | 1572.06 |
| ftv90 | 7.15 | **6.90** | 11.75 | 156.41 | 1881.75 | 31.27 | 26.82 |
| kro124p | **14.01** | 25.25 | 17.05 | 109.40 | 3305.85 | 261.86 | 215.86 |
| ftv100 | **10.60** | 16.23 | 27.68 | 133.81 | 3301.25 | 65.84 | 61.24 |
| balas108 | **186.13** | 14.18% | 14.21% | 4.91% | 23.66% | 14.32% | 14.94% |
| ftv110 | **9.91** | 38.02 | 40.40 | 201.41 | 3301.25 | 65.84 | 55.43 |
| balas120 | **1,121.96** | 10.48% | 5.55% | 12.35% | 24.38% | 7.99% | 14.22% |
| ftv120 | **35.48** | 124.80 | 131.86 | 283.99 | 2,711.11 | 385.75 | 428.64 |
| ftv130 | 26.30 | 36.18 | **13.87** | 328.68 | 340.39 | 117.30 | 97.14 |
| ftv140 | 21.98 | **20.09** | 38.30 | 535.45 | 2,120.48 | 128.32 | 93.63 |
| ftv150 | 56.25 | 37.86 | **13.47** | 561.91 | 1,142.76 | 140.09 | 104.61 |
| balas160 | **0.96%** | 7.28% | 5.97% | 11.06% | – | 7.44% | 9.24% |
| ftv160 | 93.63 | 141.24 | **91.70** | 1,347.48 | 799.79 | 288.28 | 378.75 |
| ftv170 | **57.95** | 59.31 | 66.36 | 945.09 | – | 137.08 | 172.27 |
| balas200 | **3.69%** | 14.56% | 21.31% | 24.87% | – | 27.18% | 33.09% |
| rbg323 | **112.68** | 115.78 | 176.96 | 3,274.13 | – | 141.88 | 227.59 |
| rbg358 | 255.05 | 255.62 | **131.81** | – | – | 407.22 | 867.85 |
| rbg403 | 344.33 | 335.31 | **311.92** | – | – | – | 739.97 |
| rbg443 | 292.28 | 290.01 | **282.20** | – | – | 1,050.59 | 878.86 |
| *#Minimum* | **23** | 3 | 6 | 0 | 0 | 0 | 0 |

in bold in each instance, and "-" denote that either the model runs out of memory or it is unable to obtain a feasible solution. $MTZ^{acc}$ reached optimal solutions in 30 out of the 32 cases achieving optimality gaps only in balas160 (0.96%) and balas200 (3.69%), while *MTZ*, *DL*, *GG*, *SD*, *SAWIK_en*, and *SAWIK_en−* obtained optimal solutions in 26, 26, 25, 22, 26, and 27 cases, respectively. Overall instances, $MTZ^{acc}$ reached either minimum time or gap in 23 (72%) out of the 32 instances, which is shown in the last row (# Minimum). The most significant improvement of $MTZ^{acc}$ was achieved in instance *balas*108 (solved to optimally) for which the other formulations obtained gaps over 4%. $MTZ^{acc}$ also improved the results on instance *balas*200 (solved with a gap of 3.69%), while the other formulations achieved gaps over 14.56%. Therefore, the proposed approach improves the performance of the *MTZ* model and outperform the other formulations.

We also study the impact of the proposed approach in the other MTZ-based formulations. We observe that the *DL*, *SAWIK_en*, and *SAWIK_en−* formulations generate fractional islands. Flow-based models (*GG* and *SD*) prohibit fractional islands. We denote the accelerated models by $DL^{acc}$, $SAWIK_en^{acc}$, and $SAWIK_en-^{acc}$. Results are presented in Table 4, where its columns are identical to those of Table 3. Overall, we observe that by adding valid inequalities, these formulations are also accelerated. $DL^{acc}$, $SAWIK_en^{acc}$, and $SAWIK_en-^{acc}$ outperformed their original versions in almost all instances. We have highlighted in bold for each pair of formulations (accelerated and original) the minimum time or gap. The last

**Table 4**
Comparison of the results obtained by the accelerated formulations $MTZ^{acc}$, $DL^{acc}$, $SAWIK_en^{acc}$, and $SAWIK_en-^{acc}$ with those obtained by the original *MTZ*, *DL*, *SAWIK_en*, and *SAWIK_en−* formulations.

| *Instance* | $MTZ^{acc}$ | *MTZ* | $DL^{acc}$ | *DL* | $SAWIK_en^{acc}$ | *SAWIK_en* | $SAWIK_en-^{acc}$ | *SAWIK_en−* |
|---|---|---|---|---|---|---|---|---|
| br17 | **0.15** | 0.57 | **0.11** | 0.53 | **0.12** | 0.57 | **0.11** | 1.67 |
| ftv33 | **0.27** | 0.53 | **0.15** | 0.28 | **0.64** | 0.80 | **0.42** | 0.93 |
| ftv35 | **0.45** | 0.53 | **0.78** | 0.89 | **0.92** | 0.93 | **0.76** | 1.06 |
| ftv38 | **0.32** | 0.63 | 0.82 | **0.70** | **0.50** | 0.59 | **0.63** | 0.82 |
| p43y | **1.18** | 1.29% | **1.65** | 0.93% | **1.82** | 0.64% | **2.84** | 0.58% |
| ftv44 | 1.11 | **0.90** | **0.84** | 1.33 | **1.19** | 1.23 | **1.42** | 1.98 |
| ftv47 | **0.80** | 1.94 | **1.14** | 3.44 | **1.43** | 4.41 | **1.59** | 3.09 |
| ry48p | **1.01** | 14.88 | **1.12** | 10.69 | **2.38** | 28.33 | **2.00** | 66.01 |
| ft53 | **1.23** | 13.85 | **1.29** | 2.22 | **1.39** | 19.13 | **1.40** | 12.05 |
| ftv55 | **1.89** | 3.62 | **1.57** | 1.89 | **2.68** | 6.45 | **2.03** | 7.80 |
| ftv64 | **2.53** | 8.02 | **2.12** | 7.14 | **3.05** | 10.83 | **2.70** | 10.16 |
| ft70 | **2.01** | 0.10% | **1.68** | 0.05% | **1.85** | 2.98 | **2.20** | 4.34 |
| ftv70 | **3.57** | 7.99 | **3.63** | 9.69 | **6.65** | 14.46 | **4.17** | 15.65 |
| balas84 | **74.29** | 1,319.23 | **15.83** | 1102.89 | **59.59** | 1807.99 | **195.31** | 1,572.06 |
| ftv90 | 7.15 | **6.90** | **6.36** | 11.75 | **19.00** | 31.27 | **10.21** | 26.82 |
| kro124p | **14.01** | 25.25 | **13.62** | 17.05 | **39.46** | 261.86 | **44.29** | 215.86 |
| ftv100 | **10.60** | 16.23 | **24.55** | 27.68 | 76.33 | **65.84** | **60.06** | 61.24 |
| balas108 | **186.13** | 14.18% | **55.51** | 14.21% | **154.78** | 14.32% | **290.24** | 14.94% |
| ftv110 | **9.91** | 38.02 | **23.91** | 40.40 | **46.88** | 65.84 | **40.32** | 55.43 |
| balas120 | **1,121.96** | 10.48% | **2,505.48** | 5.55% | **2.64%** | 7.99% | **2,329.82** | 14.22% |
| ftv120 | **35.48** | 124.80 | **20.24** | 131.86 | **117.85** | 385.75 | **134.35** | 428.64 |
| ftv130 | **26.30** | 36.18 | **19.40** | 13.87 | 185.17 | **117.30** | 133.12 | **97.14** |
| ftv140 | 21.98 | **20.09** | **30.22** | 38.30 | **100.98** | 128.32 | 119.92 | **93.63** |
| ftv150 | 56.25 | **37.86** | 13.55 | **13.47** | **89.96** | 140.09 | 109.69 | **104.61** |
| balas160 | **0.01** | 7.28% | **1.07%** | 5.97% | **5.33%** | 7.44% | **3.74%** | 9.24% |
| ftv160 | **93.63** | 141.24 | 102.14 | **91.70** | **199.76** | 288.28 | **251.23** | 378.75 |
| ftv170 | **57.95** | 59.31 | **33.46** | 66.36 | **125.45** | 137.08 | **113.63** | 172.27 |
| balas200 | **3.69%** | 14.56% | **1.54%** | 21.31% | **11.56%** | 27.18% | **9.96%** | 33.09% |
| rbg323 | **112.68** | 115.78 | 177.04 | **176.96** | 148.12 | **141.88** | 229.11 | **227.59** |
| rbg358 | **255.05** | 255.62 | 132.65 | **131.81** | **405.91** | 407.22 | 889.32 | **867.85** |
| rbg403 | 344.33 | **335.31** | 315.43 | **311.92** | – | – | 740.48 | **739.97** |
| rbg443 | 292.28 | **290.01** | 282.36 | **282.20** | **1,037.66** | 1,050.59 | **857.12** | 878.86 |
| *#Minimum* | **26** | 6 | **24** | 8 | **28** | 3 | **26** | 6 |

row (# Minimum) depicts the number of times that the accelerated and original formulation reach either the minimum time or gap.

### 5.2. Quality of solutions obtained by the proposed approach for ATSP

In order to evaluate the quality of solutions obtained by the proposed approach, we compare our results with a state-of-the-art metaheuristic proposed by Boryczka and Szwarc (2019b), namely HSIHM3, based on 19 instances of the ATSP considered in that paper. The authors propose an improved harmony search algorithm that is shown to improve the quality of the solutions when compared with nine metaheuristics reported in the literature (Ant Colony System, Imperialist Competitive Algorithm, Artificial Bee Colony, Genetic Algorithm, Golden Ball, Adaptive Multi-Crossover Population Algorithm, Improved Discrete Bat Algorithm, local and Variable Neighborhood Search methods, and a discrete version of Water Cycle Algorithm). In order to measure algorithm quality, (Boryczka & Szwarc, 2019b) define the average error as follows:

$$\text{Average error} = \frac{\text{Average result} - \text{optimum}}{\text{optimum}} \times 100\% \qquad (20)$$

Table 5 present the average error obtained by $MTZ^{acc}$ and HSIHM3 (Boryczka & Szwarc, 2019b). In terms of solution quality, our approach outperforms the results obtained by HSIHM3 in all instances. The average error for $MTZ^{acc}$ and HSIHM3 are 0 and 5.54, respectively. Note that our approach solves all instances to optimality within 344 seconds.

**Table 5**
Average error obtained by $MTZ^{acc}$ and by HSIHM3 (Boryczka & Szwarc, 2019b).

| Instance | Average error | |
|---|---|---|
| | $MTZ^{acc}$ | HSIHM3 |
| br17 | 0 | 0 |
| ftv33 | 0 | 2.69 |
| ftv35 | 0 | 1.19 |
| ftv38 | 0 | 1.12 |
| p43y | 0 | 0 |
| ftv44 | 0 | 2.01 |
| ftv47 | 0 | 1.38 |
| ry48p | 0 | 0.64 |
| ft53 | 0 | 3.52 |
| ftv55 | 0 | 1.07 |
| ftv64 | 0 | 2.02 |
| ftv70 | 0 | 3.99 |
| kro124p | 0 | 5.85 |
| ftv170 | 0 | 23.16 |
| rbg323 | 0 | 17.3 |
| rbg358 | 0 | 22.5 |
| rbg403 | 0 | 7.01 |
| rbg443 | 0 | 7.14 |
| Average | 0 | 5.54 |

### 5.3. Results for the mATSP

As for the ATSP, we separate fractional islands and embed the identified valid inequalities (SECs) in a commercial solver such as CPLEX and use SECs (6). In our experiments, we accelerate the ranking-based *MTZ* ($MTZ^{acc}$) and *KB* ($KB^{acc}$) formulations and compare against the original *MTZ* and *KB* as well as the GG-based formulations. We consider 11 TSPLIB problems involving 84-443 nodes and vary the number of salesman ($m$) between 2 and 4; thus, resulting in a total of 33 instances. Results are presented in Table 6. $MTZ^{acc}$ and $KB^{acc}$ improved the performance of the *MTZ* and *KB* formulations and outperformed the GG-formulation. On instances derived from balas160, $MTZ^{acc}$ and $KB^{acc}$ reached optimal

**Table 6**
Comparison of the results obtained by the proposed approach ($MTZ^{acc}$ and $KB^{acc}$) with those obtained by the *MTZ*, *KB*, and *GG* formulations on 33 mATSP instances.

| *Instance* | *m* | $MTZ^{acc}$ | *MTZ* | $KB^{acc}$ | *KB* | *GG* |
|---|---|---|---|---|---|---|
| balas84 | 2 | 43.66 | 587.66 | **11.86** | 263.02 | 200.55 |
| | 3 | 57.08 | 743.72 | **10.64** | 182.78 | 273.36 |
| | 4 | 73.17 | 241.52 | **6.75** | 101.44 | 188.05 |
| kro124p | 2 | **9.91** | 24.81 | 10.78 | 13.59 | 87.83 |
| | 3 | 13.45 | 23.03 | **7.81** | 14.34 | 93.94 |
| | 4 | 9.55 | 20.77 | **7.36** | 9.97 | 79.58 |
| balas108 | 2 | **24.30** | 13.64% | 58.59 | 13.62% | 4.99% |
| | 3 | 63.53 | 13.35% | **36.50** | 13.04% | 3.54% |
| | 4 | 66.20 | 13.16% | **35.02** | 11.24% | 1.28% |
| balas120 | 2 | **545.38** | 5.34% | 0.79% | 8.78% | 3.50% |
| | 3 | **509.09** | 5.48% | 766.94 | 5.42% | 3.03% |
| | 4 | 636.67 | 5.11% | **269.31** | 6.80% | 6.88% |
| balas160 | 2 | **3462.16** | 7.30% | 4.72% | 9.72% | 10.96% |
| | 3 | **2725.89** | 6.45% | 1.52% | 2.71% | 12.04% |
| | 4 | **2737.28** | 5.56% | 1.73% | 5.63% | 10.40% |
| ftv170 | 2 | **23.00** | 93.92 | 30.27 | 64.11 | 1139.13 |
| | 3 | 45.20 | 266.77 | **35.73** | 60.36 | 703.58 |
| | 4 | 59.27 | 56.50 | **32.69** | 76.78 | 587.44 |
| balas200 | 2 | **0.57%** | 14.27% | 4.36% | 16.71% | 19.43% |
| | 3 | **0.41%** | 17.19% | 1.94% | 18.89% | 14.01% |
| | 4 | 0.86% | 11.73% | **0.82%** | 13.58% | 24.87% |
| rbg323 | 2 | 73.06 | 72.36 | **71.47** | 73.00 | – |
| | 3 | 100.55 | 98.44 | 80.02 | **78.97** | – |
| | 4 | 64.94 | **64.77** | 314.98 | 310.89 | – |
| rbg358 | 2 | 164.39 | 162.23 | **99.59** | 100.56 | – |
| | 3 | 150.48 | 150.73 | **101.61** | 102.38 | – |
| | 4 | **102.97** | 104.67 | 141.05 | 141.97 | – |
| rbg403 | 2 | **158.22** | 160.25 | 303.83 | 306.80 | – |
| | 3 | 560.38 | 556.45 | **263.69** | 264.05 | – |
| | 4 | **172.34** | 175.19 | 244.94 | 246.17 | – |
| rbg443 | 2 | 525.50 | 522.53 | **332.58** | 338.61 | – |
| | 3 | 254.28 | **253.66** | 367.42 | 380.09 | – |
| | 4 | 375.38 | **373.84** | 389.20 | 392.17 | – |
| *#Minimum* | | 13 | 3 | **16** | 1 | 0 |

**Table 7**
Comparison of the best lower bounds (BLB) and optimality gaps (GAP) obtained by the accelerated and original formulations.

| *Instance* | $MTZ^{acc}$ | | *MTZ* | | $DL^{acc}$ | | *DL* | | $SAWIK_en^{acc}$ | | *SAWIK_en* | | $SAWIK_en-^{acc}$ | | *SAWIK_en−* | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | BLB | GAP | BLB | GAP | BLB | GAP | BLB | GAP | BLB | GAP | BLB | GAP | BLB | GAP | BLB | GAP |
| p43y | **5620.0** | 0.0% | 5,547.5 | 1.3% | **5620.0** | 0.0% | 5568.0 | 0.9% | **5620.0** | 0.0% | 5584.2 | 0.6% | **5620.0** | 0.0% | 5587.5 | 0.6% |
| balas108 | **152.0** | 0.0% | 131.3 | 14.2% | **152.0** | 0.0% | 131.3 | 14.2% | **152.0** | 0.0% | 131.1 | 14.3% | **152.0** | 0.0% | 131.0 | 14.9% |
| balas120 | **286.0** | 0.0% | 271.3 | 10.5% | **286.0** | 0.0% | 273.0 | 5.6% | **282.3** | 2.6% | 270.5 | 8.0% | **286.0** | 0.0% | 268.5 | 14.2% |
| balas160 | **393.0** | 1.0% | 384.8 | 7.3% | **392.7** | 1.1% | 383.6 | 6.0% | **390.0** | 5.3% | 381.3 | 7.4% | **389.8** | 3.7% | 382.1 | 9.2% |
| ftv170 | **2755.0** | 0.0% | **2755.0** | 0.0% | **2755.0** | 0.0% | **2755.0** | 0.0% | **2755.0** | 0.0% | **2755.0** | 0.0% | **2755.0** | 0.0% | **2755.0** | 0.0% |
| balas200 | **397.8** | 3.7% | 374.2 | 14.6% | **399.7** | 1.5% | 367.5 | 21.3% | **394.4** | 11.6% | 367.8 | 27.2% | **394.4** | 10.0% | 364.7 | 33.1% |
| rbg323 | **1326.0** | 0.0% | **1326.0** | 0.0% | **1326.0** | 0.0% | **1326.0** | 0.0% | **1326.0** | 0.0% | **1326.0** | 0.0% | **1326.0** | 0.0% | **1326.0** | 0.0% |
| rbg358 | **1163.0** | 0.0% | **1163.0** | 0.0% | **1163.0** | 0.0% | **1163.0** | 0.0% | **1163.0** | 0.0% | **1163.0** | 0.0% | **1163.0** | 0.0% | **1163.0** | 0.0% |
| rbg403 | **2465.0** | 0.0% | **2465.0** | 0.0% | **2465.0** | 0.0% | **2465.0** | 0.0% | – | – | – | – | **2465.0** | 0.0% | **2465.0** | 0.0% |
| rbg443 | **2720.0** | 0.0% | **2720.0** | 0.0% | **2720.0** | 0.0% | **2720.0** | 0.0% | **2720.0** | 0.0% | **2720.0** | 0.0% | **2720.0** | 0.0% | **2720.0** | 0.0% |
| *#Minimum* | **10** | | 5 | | **10** | | 5 | | **9** | | 4 | | **10** | | 5 | |

solutions, while all other formulations were unable to prove optimality. Also, on instance derived from balas200, we observed the benefit of accelerating the formulations by reducing the gaps considerably. The last row (# Minimum) display the number of overall instances solve in either minimum time or gap. Overall, $KB^{acc}$ outperforms the other models. Thus, our approach is also effective to accelerate mATSP formulations that use MTZ-SECs.

### 5.4. Discussion

Contrary to the previous ideas presented in the literature to enhance the MTZ-SECs, in this article, we propose an easy to implement algorithmic approach generating valid inequalities from fractional solutions while solving the original formulation. Our approach uses the values of the linear relaxation of the MTZ-ATSP formulation to generate and embed strong valid inequalities in order to improve its convergence. The proposed approach can be used by practitioners to solve practical problems, and by researchers working on metaheuristic to compare their performance against our approach on ATSP and mATSP instances in which optimal solutions are not at hand.

The effects of introducing valid inequalities on the ranking-based formulations (*MTZ*, *DL*, *SAWIK_en*, and *SAWIK_en−*) are presented in Tables 7 and 8. In the former table, we report the best lower bound (BLB) and gap obtained by the accelerated and original formulations considering the most difficult instances (with percentage gaps or unsolved by a formulation) presented in Table 3. We observe that the valid inequalities provide tighter best lower bounds than those generated by the original models, and they also improve the percentage gaps. We report in Table 8 the number of branch-and-bound (B&B) nodes explored by the accelerated and original models. We also present for the accelerated models the

**Table 8**
Number of branch-and-bound nodes explored by the accelerated and original formulations. The number of valid inequalities generated by the proposed approach is displayed by the number inside parenthesis.

| *Instance* | $MTZ^{acc}$ | | *MTZ* | $DL^{acc}$ | | *DL* | $SAWIK_en^{acc}$ | | *SAWIK_en* | $SAWIK_en-^{acc}$ | | *SAWIK_en−* |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| p43y | **385** | (98) | 6,155,220 | **496** | (83) | 8,007,059 | **1122** | (336) | 4,260,271 | **1991** | (839) | 4,260,419 |
| balas108 | **7496** | (1104) | 403,682 | **4572** | (814) | 450,356 | **5697** | (839) | 205,378 | **7478** | (688) | 194,537 |
| balas120 | **49,241** | (1753) | 170,509 | **126,907** | (2404) | 206,747 | **68,305** | (1818) | 88,917 | **60,415** | (1,986) | 105,855 |
| balas160 | **57,884** | (2666) | 88,178 | **101,695** | (2375) | 103,962 | **28,763** | (1809) | 35,827 | **24,820** | (1,124) | 40,361 |
| ftv170 | 5049 | (579) | **4010** | **1129** | (432) | 5311 | **2461** | (1034) | 6547 | **1817** | (694) | 15,868 |
| balas200 | **31,878** | (4103) | 54,051 | **51,980** | (4636) | 52,220 | **18,643** | (1988) | 27,937 | **17,570** | (2,647) | 35,107 |
| rbg323 | **0** | (0) | **0** | **0** | (0) | **0** | **0** | (0) | **0** | **0** | (0) | **0** |
| rbg358 | **0** | (0) | **0** | **0** | (0) | **0** | **0** | (0) | **0** | **0** | (0) | **0** |
| rbg403 | **0** | (0) | **0** | **0** | (0) | **0** | – | | – | **0** | (0) | **0** |
| rbg443 | **0** | (0) | **0** | **0** | (0) | **0** | **0** | (0) | **0** | **0** | (0) | **0** |
| *#Minimum* | **9** | | 5 | **10** | | 4 | **9** | | 3 | **10** | | 4 |

number of valid inequalities generated is displayed by the number inside parenthesis. We observe that the accelerated formulations reduce the number of nodes explored considerably. In the instance p43y, the number of B&B nodes explored by $MTZ^{acc}$ and *MTZ* were 385, and 6,155,220, respectively, a notable reduction in the size of the branch-and-bound tree achieved by our simple yet effective algorithmic approach.

From the computation experiments, we can observe that our approach enhances considerably the performance of MTZ-based formulations reported in the literature. This improvement is explained for the following reasons. By adding valid inequalities (facet-defining inequalities for the ATSP-polytope), the MIP formulation is strengthened, resulting in stronger linear programming (LP) relaxations. These valid inequalities cut off fractional and integer infeasible points of the LP relaxation of the MIP model. Stronger LP relaxations allow fathoming nodes without further ramification, making the enumerative solution scheme, based on the relaxation of the linear programming relaxation, more efficiently. Thus, the number of nodes in the branch-and-bound tree is drastically reduced.

It was the primary goal of this study to propose an efficient and easy to implement algorithmic approach that improves the computational performance of the MTZ models. However, we have only considered one class of valid inequalities (fractional islands), and it may be possible to embed other well-known valid inequalities reported in the literature for ATSP and mATSP to improve further the convergence of the proposed approach.

## 6. Conclusions

We have presented an easy to implement algorithmic approach that improves the performance of the Miller-Tucker-Zemlin formulation for the asymmetric traveling salesman problem by generating valid inequalities while solving the original formulation using a state-of-the-art MIP solver. We also show that our approach can be extended to accelerate other well-known ATSP models reported in the literature, and can be used to enhance ranking-based models reported for the multiple asymmetric traveling salesman problem. Computational experiments show that the proposed approach enhances considerably the performance of MTZ-based formulations reported in the literature. This algorithmic approach may be useful to solve a variety of routing problems that use MTZ-type of subtour elimination constraints.

An interesting topic for future research is to implement the proposed approach in a general hybrid metaheuristic known in the literature as Generate-and-Solve to tackle large-sized instances of ATSP and mATSP. A metaheuristic is used to generate reduced sub-instances of the original problem instance; and then, these instances can be solved by an exact solver to provide a feasible solution to the original problem. Another line of research can be to extend the proposed approach discussed in this paper to other scheduling problems, and routing problems such as capacitated vehicle routing problem (CVRP) or the precedence constrained traveling salesman problem (PCTSP), among others. Another idea is to add well-known valid inequalities reported in the literature for ATSP and mATSP, and potentially improve the convergence of the proposed approach. Finally, it is worth exploring new strong compact formulations for ATSP based on MTZ-formulation that avoid fractional islands using additional variables and/or constraints.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Credit authorship contribution statement

**Giovanni Campuzano:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing - review & editing. **Carlos Obreque:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing - review & editing. **Maichel M. Aguayo:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing - review & editing.

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